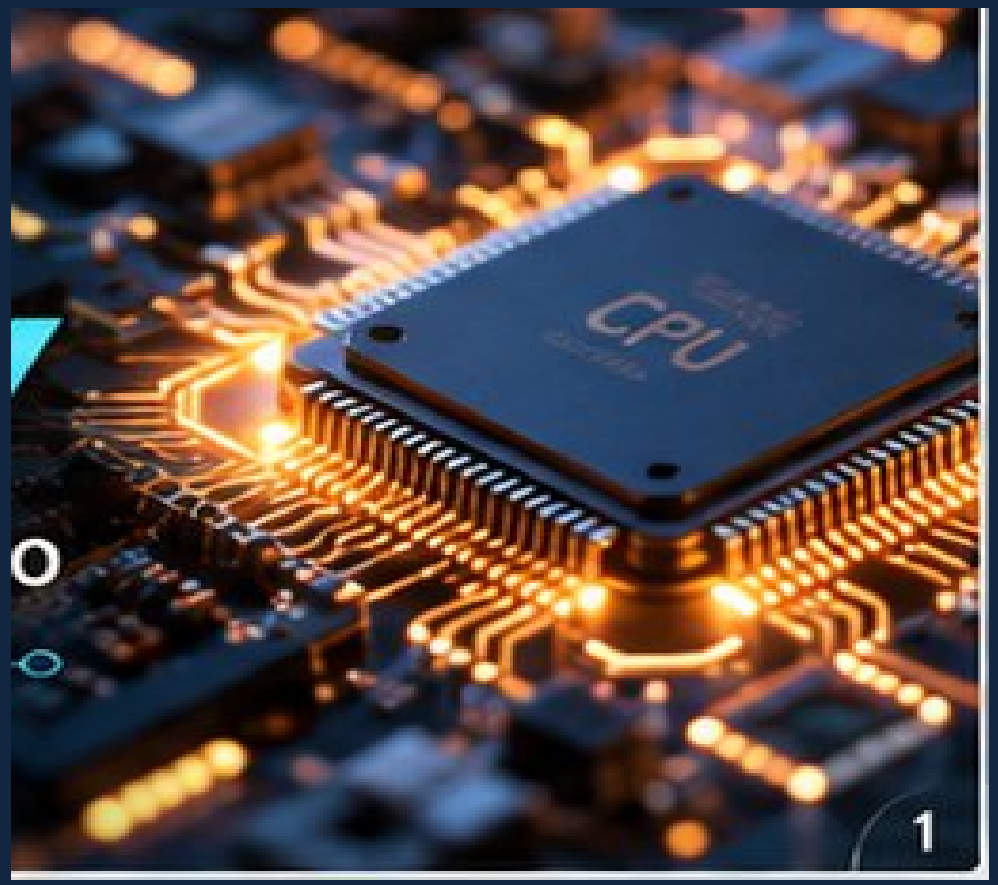
COMPUTER ORGANIZATION

CACHE MEMORY

Hit Ratio / Miss Ratio

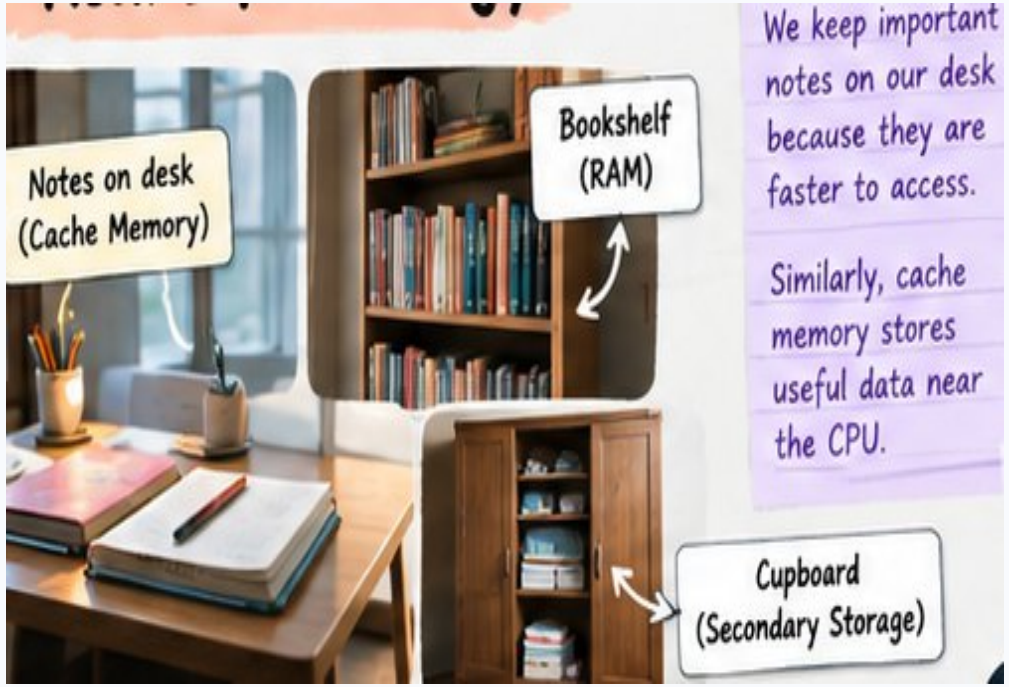
Understanding how a computer finds data quickly—and what happens when it does not.



FAST ACCESS • FEWER SLOW MEMORY TRIPS

What is cache memory?

Cache keeps useful information close to the CPU so it can be accessed quickly.



CACHE

Small, high-speed memory

STORES

Recently or frequently used data

PURPOSE

Reduce the time spent waiting for data

Think of cache as the notes on your desk: easy to reach when you need them.

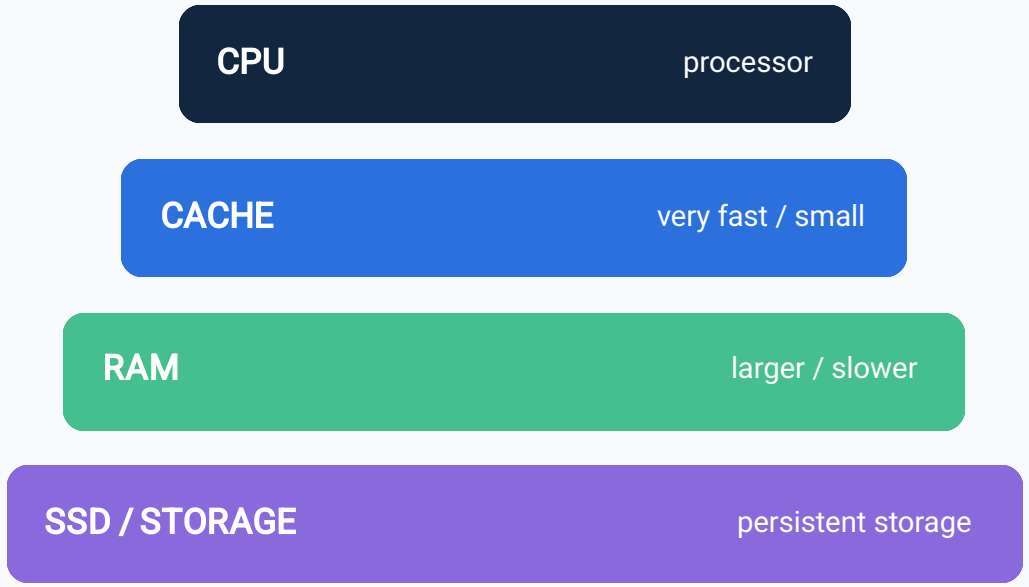
Why do computers need cache?

The processor and main memory do not operate at the same speed.

- The CPU repeatedly needs instructions and data while running programs.
- RAM is larger, but access generally takes longer than a cache hit.
- Cache keeps selected data near the CPU to reduce average access time.

Aim: less waiting, smoother processing

SIMPLIFIED MEMORY HIERARCHY



Cache can have multiple levels

Many modern processors use L1, L2 and L3 caches.

L1

Closest to the CPU

Usually the smallest and fastest cache level.

L2

A larger next level

Often larger than L1, with somewhat higher latency.

L3

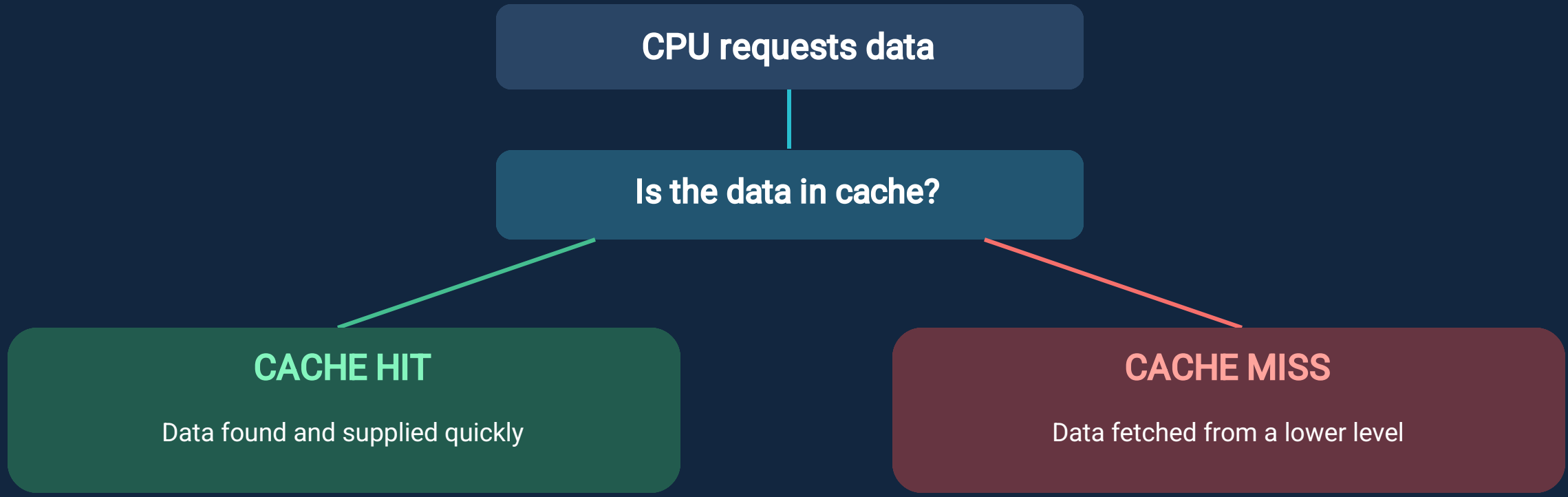
Shared on many CPUs

Often larger and may be shared by CPU cores.

Exact cache sizes and sharing differ by processor model; these are general patterns, not fixed rules.

Every request takes one of two paths

The CPU checks the cache first.



If lower-level cache also misses, the request may go to RAM.

How often is the data already there?

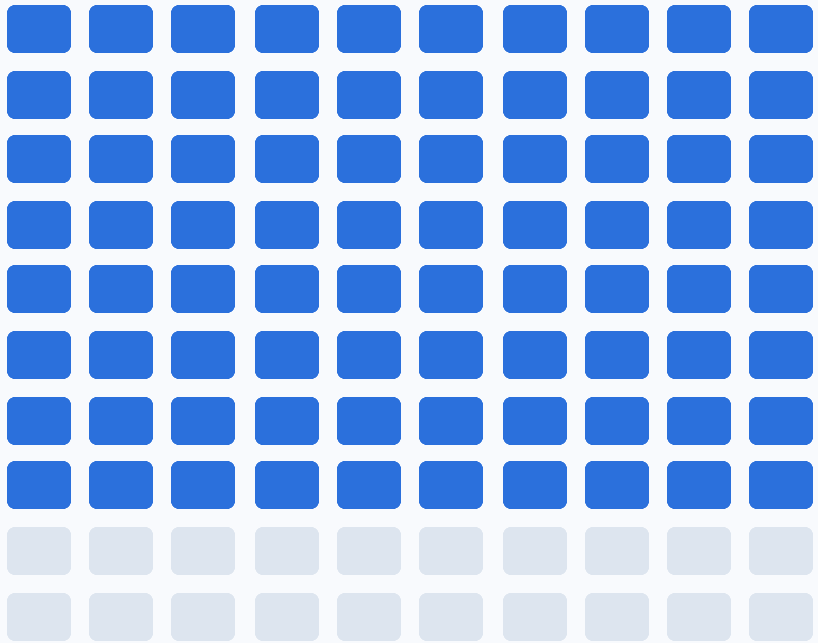
A cache hit occurs when the requested data is found in cache.

HIT RATIO

Hits ÷ Total requests × 100

- A high hit ratio means many requests are served by cache.
- It is a useful measure of cache effectiveness.

80 HITS OUT OF 100 REQUESTS



80 ÷ 100 × 100 = 80%

How often is the data absent?

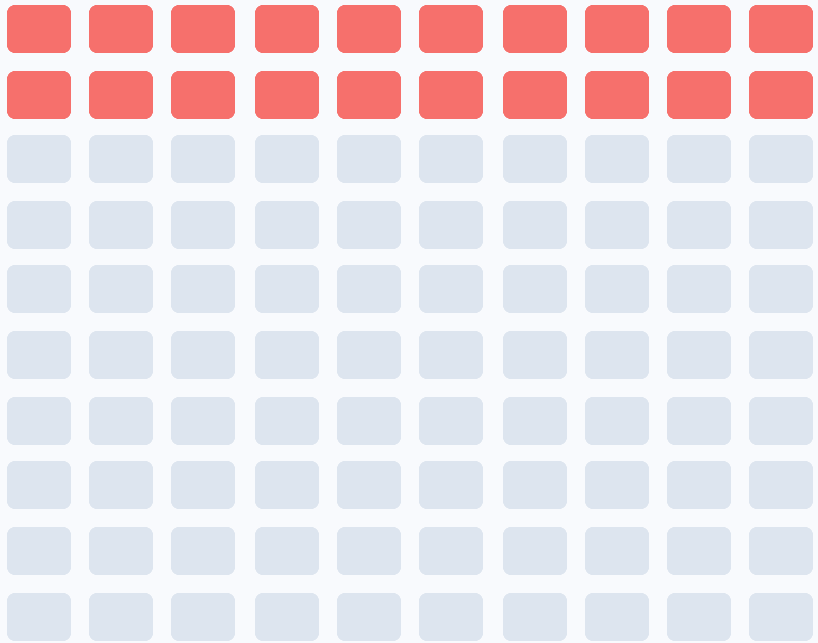
A cache miss occurs when the requested data is not found in cache.

MISS RATIO

Misses ÷ Total requests × 100

- A miss generally requires another memory level to be checked.
- Misses often increase average access time.

20 MISSES OUT OF 100 REQUESTS



20 ÷ 100 × 100 = 20%

Let's calculate both ratios

Suppose a program makes 100 memory requests.

TOTAL REQUESTS	HITS	MISSES
100	80	20

Hit ratio = 80 ÷ 100 × 100 = 80%

Miss ratio = 20 ÷ 100 × 100 = 20%

80% + 20% = 100%

Check: hits + misses must equal total requests.

The two ratios are complementary

For a single set of requests, every request is counted as a hit or a miss.



Hit ratio + Miss ratio = 100%

- Higher hit ratio generally means fewer slow-memory accesses.
- Lower miss ratio generally helps reduce average memory access time.

Why do these ratios matter?

Misses can add delay, so hit rate is only part of the performance story.

HIGHER HIT RATIO

Usually fewer expensive misses

HIGHER MISS RATIO

Usually more waiting for data

Average Memory Access Time (AMAT)

$$AMAT = \text{Hit time} + (\text{Miss rate} \times \text{Miss penalty})$$

Miss rate is the miss ratio as a decimal. Miss penalty is the extra time needed to handle a miss.

Three things to remember

Use these points to finish your presentation clearly.

CACHE

Small, fast memory close to the CPU.

HIT RATIO

How often the requested data is found.

MISS RATIO

How often the requested data is not found.

More hits + fewer misses = generally better performance

Thank you • Questions?